Which M4 Features Require Kernel Changes?

Baseline: Pyra kernel 5.6.19-pyradef-m4test-m4clean (v6) — before RPMsg integration

Assessment status: CPU0-only remoteproc with SRAM firmware and verified mailbox USER2 / IRQ34 one-shot. No RPMsg Virtio transport, no validated DDR vrings, and no second M4 core enabled.

Planned M4 Feature	Kernel Change Required?	Rationale
RPMsg commands and responses	**Yes**	RPMsg is not integrated in v6. RPMsg/Virtio drivers, resource table, vrings, DDR/IOMMU access, and mailbox kicks must first be made operational.
M4 as a compute coprocessor	**Conditional**	Small tasks are possible in principle using a custom mailbox protocol. Convenient data exchange over RPMsg requires kernel extensions.
DSP algorithms without NEON on the M4	**No**	Pure computations within M4 SRAM require no kernel changes; supplying input data and returning results must be addressed separately.
Processing sensor readings	**Usually no**	Provided Linux supplies measurements over an existing, controlled communication path. Direct M4 access to sensors may require resource reassignment.
M4-local timers and interrupts	**Usually no**	Local timers and already routed M4 interrupts can be used in firmware; new SoC interrupt sources may require changes.
Larger shared-memory regions	**Yes, likely**	v6 has no validated shared RPMsg DDR memory. Reservation, IOMMU/AMMU mappings, and cache coherency must be resolved.
DMA between M4 and DDR	**Yes, likely**	DMA channels, address spaces, buffer management, cache coherency, and ownership relative to Linux must be coordinated.
Direct I^2C, SPI, or UART access	**Depends on peripheral**	Linux drivers and the M4 must not use the same hardware without coordination; device tree, driver binding, or resource ownership may change.
M4 as an always-on background processor	**Possibly**	Operation across suspend/resume requires verification of power management, clocks, reset, and recovery behavior.
Enable the second Cortex-M4	**Yes**	v6 starts CPU0 only and holds CPU1 in reset. CPU1 startup, memory partitioning, and synchronization require new implementation and validation.
Crash detection and automatic restart	**Yes, likely**	The current safe procedure permits only one M4 start per Linux boot. Recovery requires a validated stop/reset/restart sequence.

Note: This assessment applies to the v6 baseline before RPMsg. “No” does not imply that a complete communication channel to Linux applications already exists. The IRQ34 one-shot confirms interrupt delivery, not an operational messaging transport.